

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Transformations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

TRANSFORMATIONS · CH5 · L15

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The Map

This lesson collapses Transformations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Identify the transformation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use centre and scale factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Write the vector translation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Describe a combined movement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use Transformations tag correct

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 10 website questions, 10 checked solutions, 7 local PDFs read.

01 Identify the transformation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{new point} = (1,2) + \frac{1}{2}(\text{old point} - (1,2))$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q8

Use

$$\text{new point} = (1,2) + \frac{1}{2}(\text{old point} - (1,2))$$

The vertices

$$(7,10), (15,10), (15,4), (11,4)$$

Finish: draw the enlarged shape with vertices $(4,6), (8,6), (8,3), (6,3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use centre and scale factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{new point} = (1,2) + \frac{1}{2} (\text{old point} - (1,2))$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

Source pattern: Jan 2020 P1H Q8

Use

$$\text{new point} = (1,2) + \frac{1}{2} (\text{old point} - (1,2))$$

The vertices

$(7,10), (15,10), (15,4), (11,4)$

Finish: draw the enlarged shape with vertices $(4,6), (8,6), (8,3), (6,3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Write the vector translation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 3

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q1

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale factor is

3

Finish: (a) enlargement, scale factor 3, centre (3,0); (b) vertices (1,4), (2,4), (1,6).

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

Answer ALL TWENTY FIVE questions. Write your answers in the spaces provided.
You must write down all the stages in your working. 1 O x y 10 8 6 4 2 A B 2 4 6 8 10 12
14 (a) Describe fully the single transformation that maps triangle A onto triangle B (3)
(b) On the grid above, translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ Label your triangle C (1)

YOUR SOLUTION

04 Describe a combined movement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(x, y) \mapsto (-6 - x, 4 - y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q5

(a) For a rotation about

$$(x, y) \mapsto (-6 - x, 4 - y)$$

The image has vertices

$$(-3, 9), (-9, 9), (-9, 6), (-6, 6)$$

Finish: (a) vertices $(-3, 9), (-9, 9), (-9, 6), (-6, 6)$; (b) vertices $(7, 3), (13, 3), (13, 6), (10, 6)$; (c) enlargement, scale factor 2, centre $(-3, 3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 Here is a shape P drawn on a grid of squares. x y 2 4 6 8 2 4 6 8 10 12 14 O -2 -4 -6
-8 -2 -4 -6 -8 -10 P 10 (a) On the grid, rotate shape P 180 degrees about the point (-3, 2)
Label the new shape Q. (2) (b) On the grid, translate shape P by the vector 10 8 Label
the new shape R. (1) Here are triangle T and triangle U drawn on a grid of squares.
x y 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9 10 11 12 O -1 -2 -3 -4 T U (c) Describe fully the single
transformation that maps triangle T onto triangle U. (3)

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(x, y) \mapsto (-6 - x, 4 - y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q5

(a) For a rotation about

$$(x, y) \mapsto (-6 - x, 4 - y)$$

The image has vertices

$$(-3, 9), (-9, 9), (-9, 6), (-6, 6)$$

Finish: (a) vertices $(-3, 9), (-9, 9), (-9, 6), (-6, 6)$; (b) vertices $(7, 3), (13, 3), (13, 6), (10, 6)$; (c) enlargement, scale factor 2, centre $(-3, 3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

5 Here is a shape P drawn on a grid of squares. x y 2 4 6 8 2 4 6 8 10 12 14 O -2 -4 -6
-8 -2 -4 -6 -8 -10 P 10 (a) On the grid, rotate shape P 180 degrees about the point (-3, 2)
Label the new shape Q. (2) (b) On the grid, translate shape P by the vector 10 8 Label
the new shape R. (1) Here are triangle T and triangle U drawn on a grid of squares.
x y 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9 10 11 12 O -1 -2 -3 -4 T U (c) Describe fully the single
transformation that maps triangle T onto triangle U. (3)

YOUR SOLUTION

06 Use Transformations tag correct

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 3

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q1

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale factor is

3

Finish: (a) enlargement, scale factor 3, centre $(3, 0)$; (b) vertices $(1, 4)$, $(2, 4)$, $(1, 6)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

Answer ALL TWENTY FIVE questions. Write your answers in the spaces provided.
You must write down all the stages in your working. 1 O x y 10 8 6 4 2 A B 2 4 6 8 10 12
14 (a) Describe fully the single transformation that maps triangle A onto triangle B (3)
(b) On the grid above, translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ Label your triangle C (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Identify the transformation
"work out", "show", "hence", a familiar exam phrase	Use centre and scale factor
"work out", "show", "hence", a familiar exam phrase	Write the vector translation
"work out", "show", "hence", a familiar exam phrase	Describe a combined movement
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use Transformations tag correct

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.